

$$S \rightarrow ABA$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

1) Eliminate ϵ -productions

$$A \rightarrow \epsilon$$

$\therefore A, B$ are nullable

$$B \rightarrow \epsilon$$

$$S \rightarrow ABA$$

$$S \rightarrow ABA \mid BA \mid AA \mid AB \mid A \mid B \mid \epsilon$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$\rightarrow$ S has ϵ so add new symbol

$$S_0 \rightarrow S \mid \epsilon$$

step 3)

$$\cdot S \rightarrow A$$

$$\cdot S \rightarrow B$$

$$\cdot S_0 \rightarrow S$$

thus

$$S_0 \rightarrow ABA \mid BA \mid AA \mid AB \mid aA \mid a \mid bB \mid b \mid \epsilon$$

$$S \rightarrow ABA \mid BA \mid AA \mid AB \mid aA \mid a \mid bB \mid b$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

step 4) Replace terminal

$$x_i \rightarrow b$$